Problem 16.4

A 5-year annuity due has periodic cash flows of \$100 each year. Find the present value of this annuity if the effective annual interest rate is 8%.

Solution

Since this is an annuity due, payments are made at the beginning of each year. The present value is

$$PV = 100\ddot{a}_{\overline{5}|}.$$

For an annuity due,

$$\ddot{a}_{\overline{n}|} = (1+i)a_{\overline{n}|}.$$

Using $i = 0.08$,

$$\ddot{a}_{\overline{5}|} = (1.08)\left(\frac{1-(1.08)^{-5}}{0.08}\right).$$

Thus,

$$\begin{aligned} PV &= 100(1.08)\left(\frac{1-(1.08)^{-5}}{0.08}\right) \\ &\approx 431.2127. \end{aligned}$$

Hence, the present value is

$$\boxed{\$431.21}.$$